

Project Proposal — Padel Predictor: pre-match win probabilities for the Premier Padel / FIP tour

Noah Rodriguez · github.com/Noah-Rod/padel-predictor (public)

1 Problem statement

For every scheduled **main-draw match** on the **Premier Padel** tour (Major, P1, P2, Finals) and, where the feed covers them, the **FIP Platinum and Gold** tiers, men's and women's draws, predict the **probability that team A beats team B**. Predictions are produced **daily for all fixtures up to seven days ahead**, i.e. from the moment a draw is published until the match starts, and are served to **fans and fantasy-league players** who fill in brackets before play begins.

Scope: the 2026/27 season live, backfilled with completed draws from January 2023. Qualifying rounds and Silver, Rise and Promotion events are excluded: too many players with no history.

Success criterion, on an out-of-time test set (the most recent 20 % of matches, cut at a tournament boundary), against three baselines:

- **rank favourite** (the team with more FIP ranking points wins): accuracy must be higher, and $\geq 70\%$ in absolute terms;
- **replayed Elo** ($P(A) = 1 / (1 + 10^{((R_B - R_A)/400)})$ from the same in-house player ratings): log loss must be lower;
- **the data provider's own Elo simulation** (padelapi.org publishes a per-player elo and a win-probability formula that it reports at 71.8 % accuracy and 0.517 log loss on 7 500 matches): log loss must be lower on my test set. If the model only matches the provider's rating, it has added nothing.

A candidate that fails any check is not promoted; the current champion keeps serving.

2 Originality & motivation

I play padel and follow the Premier Padel tour. The useful version of this system is the one I would open on a Wednesday to pick a bracket before Thursday's round of 16.

Checked against mlops-lab.ch (FS26: 15 projects, three sport-performance forecasters for golf, high jump and kiteboarding, no head-to-head match predictor) and the KTH ID2223 showcases 2022/23 to 2025/26 (134 projects, eleven sports or game predictors: football $\times 3$, Fantasy Premier League, NHL $\times 2$, chess, Clash Royale, League of Legends $\times 2$, transfer values). None covers padel, tennis or any doubles sport. What makes this different from the tennis predictors it resembles, and from the provider's own Elo: **padel is doubles-only and pairs split and re-form every few months, so there is no stable team entity to rate**. A rating answers "how strong are these four players"; this system asks what the rating misses: how long the pair has played together (partnership), current form, rest, tier and the official ranking as it stood on match day. The bar is set by the provider's Elo, which already averages two players into a pair, so the project only succeeds if those signals carry information beyond a rating. The MLOps-side twist is a **point-in-time ranking join**: rankings and Elo are re-published weekly, and attaching today's values to a 2024 match is the leak most sports predictors quietly contain.

3 Data source & features

Source. padelapi.org (REST, JSON, bearer token; run by Fantasy Padel Tour): seasons $\rightarrow$ tournaments $\rightarrow$ matches, plus players and rankings, with consistent player IDs across Premier Padel and the FIP Tour. Matches are read per tournament with an `updated_after` filter; each carries status, draw (main / qualy), `scheduled_at`, round and the four player IDs. Free tier: 50 000 requests/month, 10/min, 2 000/day; schedules, results of the last six months and the current ranking only. The paid tier adds the full archive (complete draws for Premier Padel and FIP from 2023) and the **ranking-history endpoint** (weekly snapshots of official points and provider Elo per player, hidden on the free plan). I subscribe for the backfill, or use the provider's academic access (a request naming the university and project) if granted first. A daily sync of the running tournaments costs under 50 requests, so the live pipeline stays inside the free quota.

Update frequency and volume. The feed updates during tournaments, which run Wednesday to Sunday almost every week of the season; start times move during the day and are re-estimated hourly. Estimated 3 000 to 4 000 labelled main-draw matches per season across both categories, roughly 10 000 since 2023, growing by 60 to 120 per tournament week. Should the archive prove shallower, history starts six months before the poller and the test set grows with it: this degrades the numbers, not the system.

Fallback. Scraping the results and rankings pages of padelfip.com (same federation, same player names) behind the same MatchSource interface; its terms of use are checked before any scraping and reported in the MS2 summary.

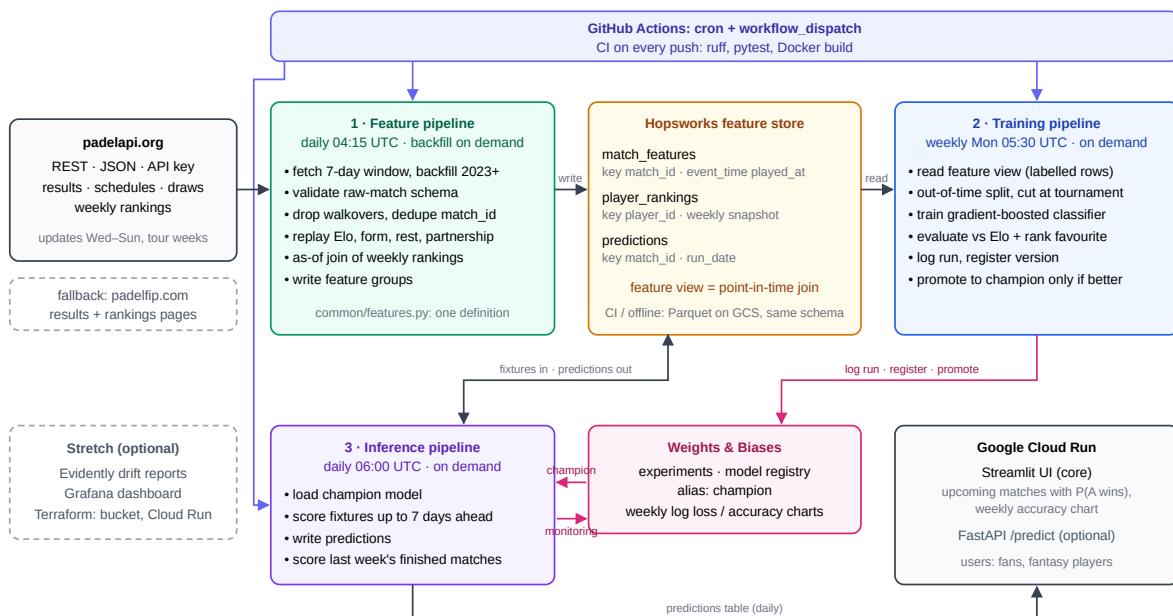
Second line: the last good feature-store snapshot plus the deterministic synthetic source that CI already runs.

Label. `team_a_won` $\in \{0, 1\}$ for matches with `status = finished`, from the source's winner field. walkover, retired and bye matches are excluded from training and evaluation (they still update rest days); scheduled fixtures carry a null label and are what inference scores. Team A / B is assigned by a hash of the pair names, never by seeding, so the label is balanced by construction ($\approx 50/50$) and no rare-class handling is needed; every row is also mirrored (swap teams, negate differences, flip label) to enforce symmetry. Nothing in the feature row is derived from the result: set scores, duration and status are ingested for filtering only and never joined into features.

Features (team A minus team B unless stated), each computed from state strictly before the match: `elo_diff` (in-house player Elo, $K = 24$, initial 1500, replayed match by match), `provider_elo_diff` (padelapi.org Elo from the last weekly snapshot before the match), `rank_points_diff` (official FIP points, same as-of rule), `form_diff` (win rate over each player's last five matches), `rest_days_diff` (days since the team's freshest player last played, capped at 60), `partnership_diff` (matches the pair has played together), `tier_level` (Major 4 ... Gold 1), `round`.

Leakage controls. Out-of-time split, never inside a tournament. Elo, form, rest and partnership state is updated only *after* a match's feature row is built; a unit test asserts that flipping a match's winner does not change its own features. Rankings and provider Elo are joined as-of the match date, never the current values. Re-ingested matches are de-duplicated on `match_id`, keeping the newest record, so a fixture that later completes becomes one labelled row.

4 System design



Three pipelines share one feature-definition module, `src/padel_predictor/common/features.py`, so training and inference cannot drift.

Layer	Choice	Why
Feature store	Hopsworks (serverless)	Feature groups with event time and a feature view for as-of joins; survives ephemeral CI runners. Parquet on GCS with the same schema stays as the CI/offline backend.
Experiment tracking & registry	Weights & Biases	Runs, metrics, model versions with a champion alias and the weekly monitoring charts in one hosted tool.
Orchestration	GitHub Actions (cron + workflow_dispatch)	Feature daily 04:15 UTC, training weekly Monday, inference daily 06:00 UTC; nothing to host.
Serving	Google Cloud Run	Streamlit UI reading the daily predictions table; scales to zero.
Monitoring	Inference pipeline → W&B	Rolling weekly log loss and accuracy on newly finished matches, shown in the UI.

Stretch, explicitly optional, only after the core is live: FastAPI `/predict` for ad-hoc matchups; Evidently drift reports on `elo_diff` and `rank_points_diff`; a Grafana dashboard; Terraform for the bucket and the Cloud Run service. The repository is public, CI runs lint and tests on every push, and milestones are tagged `ms1` to `ms4`.